

MELVIN SHAREES

Computer & Autonomous Systems Engineering Student | Cloud · Data · AI/ML · Software

Dubai, UAE • +971 50 800 8474 • melvin.s@outlook.com

linkedin.com/in/melvin-sharees • github.com/Melvinsharees

Indian • UAE Golden Visa • English (fluent)

PROFILE

Computer & Autonomous Systems Engineering student at a Distinction average, with hands-on experience building systems end to end. Led software integration on a five-person team that shipped a working offline AI wearable, owning the real-time pipeline from audio capture through speech recognition to AR output. Comfortable across Python, C/C++, and MATLAB, and equally at home in autonomous robotics and statistical analysis.

TECHNICAL SKILLS & COMPETENCIES

Programming Languages: Python, MATLAB, C (Fundamentals), C++ (Fundamentals)

Hardware Frameworks: Raspberry Pi 5 (System Design Phase), PC Assembly & Component Diagnostics

Core Engineering: Digital Signal Processing (DSP) Foundations, Digital Hardware Concepts

Software & Design Tools: Multisim, AutoCAD, Canva, Microsoft Office Suite (Word, Excel, PowerPoint), Tinker CAD

EDUCATION

University of Wollongong in Dubai (UOWD)

Expected 2028

Bachelor of Engineering (Honours) — Computer & Autonomous Systems Engineering

Weighted Average: 79/100 (Distinction average) • Completed 2 of 4 years

Relevant coursework: Programming for Engineers, Digital Hardware, Circuits & Systems, Digital Signal Processing, Engineering Statistics, Electronics.

CERTIFICATIONS & AWARDS

- CS50P: Introduction to Programming with Python — Harvard University (2026)
- Circuitry & Soldering and Intro to Robotics Workshops — UOWD (2025)
- Duke of Edinburgh's International Award — Bronze (2023)

PROJECTS

Lenso — Offline AR Speech-to-Sign-Language Wearable

Python · Raspberry Pi 5 · Whisper · NLP

Lead Software Integrator | 5-person team | Live prototype demoed at Innovation Fair

- Built the end-to-end pipeline (microphone capture → offline Whisper transcription → NLP → 3D AR avatar) on a Raspberry Pi 5, delivering real-time translation with no cloud dependency.
- Designed and built a smart wearable that translates spoken audio into a real-time sign-language avatar with subtitles.
- Integrated OpenAI Whisper for offline speech recognition, selecting and tuning a lightweight model to run in real time within the constraints of embedded hardware.

Autonomous Smart Sorter Robot

C · Webots · Computer Vision

- Engineered a fully autonomous controller for a Pioneer 3-DX mobile robot in C, using a finite-state machine to manage complex navigation logic.
- Integrated computer-vision algorithms on the camera feed to dynamically identify, track, and sort coloured objects.
- Developed robust obstacle-avoidance and targeted navigation protocols for collision-free movement and successful mission completion.

Statistical Machine Failure Analyzer

MATLAB · Data Analytics · Excel

- Conducted statistical analysis on a 1,000-observation industrial dataset to evaluate machine-failure risk from sensor data.
- Executed probability-distribution fitting, sampling-distribution analysis, and hypothesis testing to validate system operational requirements.
- Computed 95% confidence intervals and extracted actionable performance insights using MATLAB and Excel.

Vending Machine State Controller

C / C++

- Programmed state-machine logic for a vending system: dynamic inventory allocation, coin calculation, and exact change dispensation.
- Built a secure administrative mode enabling operators to replenish stock, dynamically adjust pricing, and track total sales data.

Op-Amp Function Generator with Frequency Control

Multisim · Ultiboard · PCB Design

- Designed a three-waveform signal generator (square, triangular, sine) using a Schmitt trigger, integrator, and second-order active filter, tunable from 544 Hz to 55 kHz.
- Validated through hand calculations, Multisim simulation, and breadboard testing — all seven specifications passed — then produced the PCB layout in Ultiboard with zero DRC errors.

Temperature Threshold Alarm

Digital Hardware Design

- Designed and simulated a hardware-based temperature-alarm system using LM339 comparators and a 555 timer circuit.
- Engineered proportional voltage logic with potentiometers to map simulated temperature inputs and trigger automated, state-based visual LED alerts.